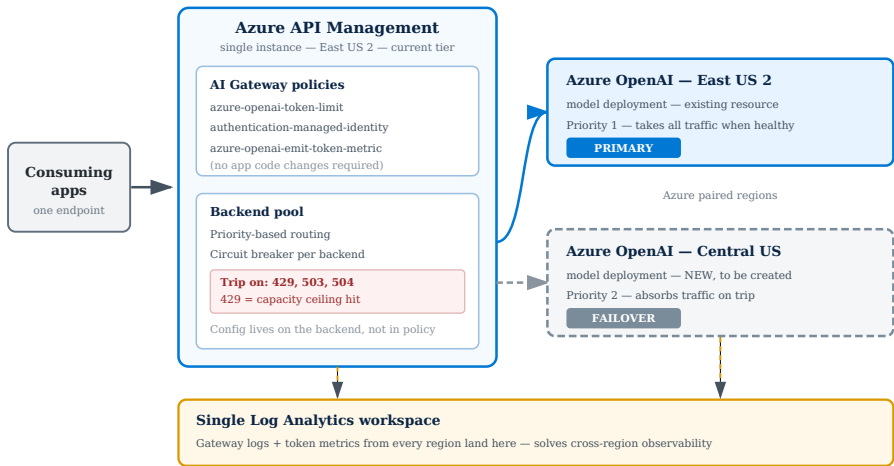


Multi-Region Azure OpenAI: Recommended Architecture

Load balancing and failover across East US 2 and Central US — options review and recommendation

Summary. There are two separate decisions here, and only one of them carries a licensing cost. Making the **Azure OpenAI backends** multi-region delivers the load balancing and failover we need, and it works on our current API Management tier. Making the **API Management gateway itself** multi-region is a separate, far more expensive decision that we can defer.



Recommended architecture — Microsoft’s “AI Gateway” pattern using API Management backend pools. — normal traffic — path taken once the circuit breaker trips

How it works

API Management exposes one stable endpoint. Each Azure OpenAI resource is registered as a *backend*, and the backends are grouped into a *pool* that distributes requests across instances in different regions.

- **Routing.** Round-robin is the default; weighted and priority-based routing are also supported, so we can express “prefer East US 2, fall back to Central US” rather than splitting traffic evenly.
- **Circuit breaker.** A property on the backend — failure count or percentage within a time window, plus the status codes that count as failures. On trip, the backend is marked unhealthy and the pool skips it until a probe succeeds. For Azure OpenAI, 429 belongs in that list alongside 503 and 504: it signals the throughput ceiling was hit, which is exactly when we want to shift regions rather than return an error.
- **No client changes.** Backend pools need no policy configuration; the rules are properties of the backend.

On the tier question: this pattern runs from a **single-region** API Management instance. Only the Azure OpenAI resources are duplicated. Premium tier is **not** required.

Options considered

Approach	What it protects against	Tier / cost impact	Assessment
A. APIM backend pool + circuit breaker Microsoft AI Gateway pattern	AOAI regional outage; throttling and capacity exhaustion in one region	No tier change. Cost is the second AOAI resource only.	RECOMMENDED Microsoft’s documented pattern. Meets both requirements and is Terraform-expressible.
B. APIM Premium multi-region	Everything in A, plus loss of the APIM gateway itself in a region	Premium (classic) only. Roughly \$2,800/unit/month, three units per region recommended for AZ coverage.	DEFER Only the gateway is replicated — management plane and developer portal stay in the primary region. Hard to justify until gateway downtime is shown to be the real risk.
C. Azure Front Door + two independent APIM instances	Same coverage as B	2× current tier + Front Door. Materially cheaper than Premium.	ALTERNATIVE TO B The budget route to gateway resilience. More moving parts: config drift between two instances becomes ours to manage.

What this requires

Component	Change required
Consuming applications	None. Same endpoint, same keys, same SDK calls.
API Management	Add a second backend, a pool, and circuit-breaker rules. Stays in one region, stays on the current tier.
Azure OpenAI	New resource and matching model deployment in Central US, with quota equal to the primary — unequal quota makes failover a brownout rather than a failover.
Terraform	New backend and pool resource definitions.
Monitoring	Shared Log Analytics workspace, with diagnostic settings wired in both regions before the second region goes live.

Sources: Microsoft Learn (API Management reliability; deploy API Management to multiple regions) and Microsoft Community Hub (Unified AI Gateway design pattern). Pricing figures are indicative list prices and should be confirmed with procurement.